df.corr()

Univariate Analysis

Suppose you have multiple columns and If independent Analysis of single column is called Univariate Analysis.

- When Analysis done on both column that is Bivariate

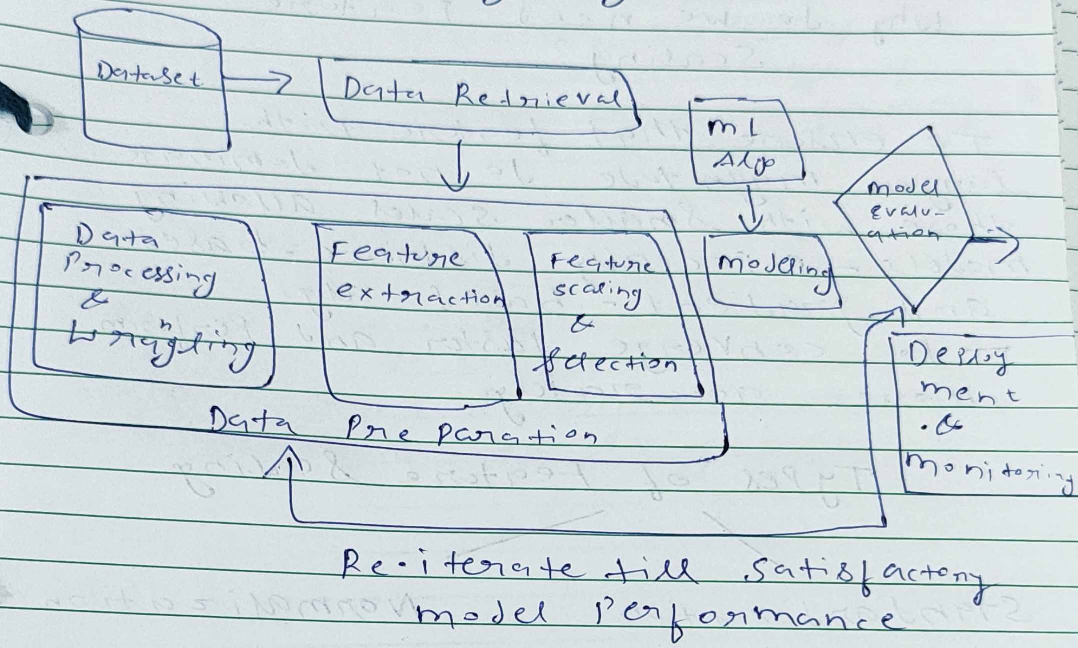
- When you run analysis on more than two ~~or~~ ~~more~~ columns that is multivariate.

data

Numerical — categorical

Feature Engineering

It is the process of using domain knowledge to extract features from raw data. These features can be used to improve the performance of machine learning algo.



Feature Engineering

- Feature Transformation
- Feature construction
- Feature selection
- Feature extraction

Feature Transformation:

- Missing values
- Handling categorical Features
- Outlier Detection
- Feature scaling

Feature Scaling

It is a technique to Standardize the independent features Present in the data in a fixed range.

Why do we need Feature Scaling?

To ensure that feature with larger magnitude do not dominate those with smaller scales, allowing models - especially distance-based and gradient-based algo - to converge faster and Perform more accurately.

Types of Feature Scaling

Standardization   Normalization

- Also called as z-score Normalization

Q When to use Standardization?

K-means	Use Euclidean distance measure
K-nearest Neighbours	measure the distance b/w pair of samples and these distance are influenced by measurement unit.

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Principal component Analysis (PCA)	Try to get feature with maximum variance
Artificial Neural Network	Apply Gradient Descent
Gradient. Descent	Theta calculation become faster after scaling and learning rate in the update equation of stochastic G.D is same for every parameter.

Normalization

Goal of Normalization is to change the values of numeric columns in the dataset to use a common scale, without distorting difference in the ranges of values or losing info.